

CWY2: direct rules without a special column

DEFINITIONS & PROOFS / ENGLISH EDITION

Definition: 2026-09-16. Repository edition: 2026-09-17; Section 7 incorporates the subsequent existing paper proof. The source notation is **omega-Y weak magma**, not weak-omega-Y. The mathematical rules do not depend on a website; NER is a resource-bounded implementation.

The released NER file has ID cwy2-direct-v2. The earlier legacy design, with [5] root splitting and different seed indices, is not included here. A legacy expansion path must not be silently reused as a current path.

1. Finite column lists

A finite expression is a sequence $G = (C_0, \dots, C_j)$ of columns. This is both the mathematical data and the default string. There is no second layer of root splitting and no special marker column. Limit is a separate external symbol.

Column j is a finite list

$$C_j = ((p_1, W_1), \dots, (p_r, W_r)), \quad j > p_1 > \dots > p_r \geq 0.$$

W is a nonempty nondecreasing word. Its letters belong to $\{0, \dots, p\} \cup \{j\}$; j denotes SELF and may be entered as $*$. Keep only one copy of the repeated first letter: $(\emptyset, \emptyset, 2, 2, 5)$ normalizes to $(\emptyset, 2, 2, 5)$. Write N for this normalization.

Words compare by first letter, length, then lexicographically on remaining letters; write $<_*$. In a column $W_1 <_* \dots <_* W_r$. The first column is empty; an empty final column is a successor suffix. Zero is the sequence with no columns, displayed as $\emptyset$, not the one-column expression $[\]$.

Ordinary columns are written $[p : (\text{word}) ; p : (\text{word})]$ and concatenated without separators. A record compares by its parent address in **usual natural-number order**, then by its word. Decreasing storage order does not reverse the parent comparison. Columns compare lexicographically by records and whole terms lexicographically by columns; proper prefixes are smaller.

The two-column seeds are

$$E_1 = ([\], [\]), \quad E_m = ([\], [(0, (0, \underbrace{1, \dots, 1}_{m-2}))]) \quad (m \geq 2).$$

They are exactly $\Phi(1, m)$ under the old highest-profile encoding, but the algorithm below is **not defined by decoding**. The standard domain is generated by corresponding seeds, the chosen expansion operations and prefixes. Freely written structurally valid tables do not automatically share the standard-domain well-ordering claim.

2. Two finite helper operations

Sky, implemented as skyline, takes finitely many records:

1. For each parent retain only its greatest word.
2. Sort parents in decreasing order.
3. Retain a row only when its word exceeds every word already retained.

Let $L_{a \rightarrow b}(W)$ replace every occurrence of a by b . Lifting a parent column changes words, not that column's own parent addresses:

$$\text{lift}_{a \rightarrow b}(C_a) = \{(r, L_{a \rightarrow b}(U)) : (r, U) \in C_a\}.$$

These are finite list operations, not huge numeric countdowns.

3. Look below the original cap

For W let q be its last letter. Define $B_G(W)$ by scanning C_q in storage order:

1. Remember the last word U_- strictly below W , or the absent value $\perp$ if none.
2. If all words are below W , return U_- .
3. Otherwise let (r, u) be the first row whose word is at least W . Replace the **first** q in W by r and normalize, obtaining V .
4. Return $\max_*(U_-, V)$, with $\perp$ below all words.

This scans one existing table. It does not recursively decrement that table or graft a new parent tail. In particular, if $W = (q)$, every word in C_q is smaller; the result is its highest word or $\perp$.

For a nonempty last column j with last row (p, w) , let

$$\begin{aligned} R_G(C_j) &= \text{Sky}((C_j \setminus \{(p, W)\}) \cup \{(p, L_{q \rightarrow j}(B_G(W)))\}), \\ D_G(C_j) &= \text{Sky}(R_G(C_j) \cup \text{lift}_{p \rightarrow j}(C_p)). \end{aligned}$$

Omit the inserted row if the query returned $\perp$. The implementation names are `removeOriginalTop` and `minusColumn`. R removes only the **original cap**; D then adds the parent list at the target column. Performing both operations in the borrowed origin column would manufacture a tail not belonging below the original cap; that was the earlier failed candidate.

Fixed regression example

The standard wY input $(1, 4, 15, 51)$ encodes as

```
[][0:(0,1,1)][1:(0,1,2)][2:(0,1,2)]
```

Here $p = q = 2$ and $W = (0, 1, 2)$. Querying the original lower profile in column 2 gives $(0, 1, 1)$. Then lifting its SELF references into column 3 gives the correct lowered last column

```
[2:(0,1,1);1:(0,1,3)]
```

This agrees with the encoding of $(1, 4, 15, 50)$. The old algorithm decremented the borrowed column itself and introduced the larger unwanted word $(0, 2, 2)$. The fix changed the queried object; it did not merely remove a failing assertion.

4. Default fundamental sequence

Zero stays zero. For a successor, every n th term deletes the last empty column. A finite limit's zeroth term also deletes its last column.

For a limit let j be the final column, c its controlling last parent, $w = j - c > 0$, and $D = D_G(C_j)$. Define

$$\phi_b(t) = \begin{cases} t, & t < c, \\ t + bw, & t \geq c. \end{cases}$$

Apply this to parents and every word address. Keep $C_0, \dots, C_{j-1}$ and, for $b = 0, \dots, n - 1$, append

$$\phi_b(D), \phi_{b+1}(C_{c+1}), \dots, \phi_{b+1}(C_{j-1}).$$

Each block has w columns. When $w = 1$, the block contains only its first column, with no extra empty column. The core `fs` uses only D , the fixed source columns and address shifts. It does not run numeric wY decrement or

construct mountain cells under another name. Storage is controlled by the actual input and output lists.

5. Top, numbering and the legacy distinction

Top, Limit and ∞ denote the external symbol displayed as **Limit of CWY2**, above every finite expression. It is not an ordinary column. Define

$$\text{Limit}[n] = E_{n+1} \quad (n \geq 0).$$

All three NER expansion options use this top rule, corresponding to wY's $(1, n+1)$. The seeds increase, but need not be prefixes of one another: **the external top is exempt from the prefix requirement**. In particular, Limit[0] is $[\][\]$, not $[\]$.

Finite terms have no special seed shift:

Input	Canonical list	wY sequence
E3	$[\][\emptyset: (\emptyset, 1)]$	1, 3
E3[0]	$[\]$	1
E3[1]	$[\][\emptyset: (\emptyset)]$	1, 2
E3[2]	$[\][\emptyset: (\emptyset)][1: (1)]$	1, 2, 4
E4[2]	$[\][\emptyset: (\emptyset, 1)][1: (1, 2)]$	1, 3, 10

Paths such as E3[2] always mean default FS. A bare natural number means that many empty columns. The parser rejects old $[S]$, $[\top]$ and S3 syntax; translate through the old numeric view instead.

Other NER conventions

Let F_n be the default term and P the retained original prefix. For finite limits:

- FS_alter appends $\phi_n(D)$ to F_n . Its zeroth term is P followed by D, so it does not satisfy the deletion rule.
- FS_short, or Inz-1, has zeroth term P and first term P followed by D. For $n \geq 2$ use F_n if $w = 1$, and F_{n-1} if $w > 1$.

Zero and successor rules are unchanged across the three options. These align separately with wY's three conventions; they are not three names for the same function. The main prefix/deletion requirements refer to default FS.

Relation with legacy

Let U undo the old root splitting. For a legacy pure two-column limit seed,

$$U(A_{\text{legacy}}[n]) = U(A_{\text{legacy}})[n+1]_{\text{new}}.$$

The new zeroth term instead deletes the last column. Other finite default indices are unchanged. The new nth top term corresponds to the old $(n+1)$ st top term after unpacking. This is not merely renaming $[S]$ and does not assert equality at identical legacy/current paths.

6. Immediate formal properties

Increasing n by one appends one whole block to the default term. The zeroth term is the retained prefix. Hence every nonzero finite A satisfies

$$A[0] = \text{delete-last-column}(A), \quad A[n] \preceq A[n+1].$$

For limits the added block is nonempty; for successors all terms coincide. There is no root-conversion or pure-seed exception. The external top is outside this assertion.

For fixed limit A, the width, source record counts, word lengths and D are fixed. Only addresses grow, bounded by $O_A(n)$ and requiring $O_A(\log(n+2))$ decimal digits. There are linearly many actual columns, so

$$\Omega_A(n) \leq |\text{str}(A[n])| \leq O_A(n \log(n+2)).$$

The nth top seed also has explicit length $\Theta(n)$. This is a theorem about full default lists, not the size or runtime of optional numeric or mountain views. Successor output is fixed and has no linear lower bound.

7. Existing paper equivalence and well-ordering transfer

The subsequent equivalence paper, written on 2026-09-16, proves the direct decrement identity, exact address-copy identity and strict order preservation:

$$f_n(\Phi(s)) = \Phi(F_n(s)), \quad s <_{\text{lex}} t \iff \Phi(s) <_{\text{collex}} \Phi(t).$$

The map preserves zero, width, prefixes, successors and corresponding seeds. It is onto the corresponding generated standard domain, giving an **order isomorphism**, not just a lower-bound embedding. Default, long and lnz-1 conventions commute separately. CWY2 therefore has the same standard order type as wY, not a proved strict increase.

Sections 2-5 of the paper supply the four obligations missing from the earlier rules note: low-node lifting, positive-dimensional intermediate layers, zero-dimensional origin inheritance, and domination of the extra lower parent rows. Section 6 gives the exact-copy argument rather than treating an old inequality as equality.

This is a **paper proof depending on the finite lemmas of [W]**, not a Lean certificate. Assuming [W]’s wY well-ordering theorem, the isomorphism transfers well-ordering in its KP_ω + there exists an uncountable ordinal framework. This release neither re-audits all of [W] nor proves the free-table order well-founded.

The NER snapshot and readable core retain their original bytes, including historical comments saying equivalence is pending. The paper and this section supersede those progress comments. No mathematical algorithm changed during packaging.

8. Views, source and usage

Import the complete standalone NER script. The readable core contains the default expansion and column comparison without reading a mountain. Only the optional display layer reconstructs numbers and geometry. A view failure does not fall back to the original wY FS.

The main list writes every address. Other views are ω -Y 数列, DBMS, DBMS_MN and ADBMS. Enable NER’s diagram option for the complete mountain. There is no redundant SELF-list view. Inputs include E3, E4[2], Top[3], natural numbers and complete lists. Time, width, word-length and numeric-view limits are runtime protections, not mathematical rules or nontermination verdicts.

```
import {seed, fs, display} from './cwy_direct.mjs';
console.log(display(fs(seed(3), 2)));
// [[0:(0)]:[1:(1)]]
```

[W]: user-supplied *Well-foundedness of the Weak-Magma omega-Y System in KP with an Uncountable Ordinal* (omega-Y-Weak-Magma-KP.pdf), not redistributed. Dependencies: equivalence paper. Code and diagram credits: Yukito/Naruyoko and NER; see SOURCES. Bounded checks, not proofs: VALIDATION.